

MUSTAFA AL LAIL

School of Engineering, Texas A&M International University
5201 University Boulevard, Lamar Bruni Vergara Science Center 324-E, Laredo, TX 78041
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EDUCATION

Colorado State University, Fort Collins, Colorado **August 2018**
Doctor of Philosophy in Computer Science
GPA: 3.9 out of 4.0
Advisers: Professor Robert B. France and Professor Indrakshi Ray

Colorado State University, Fort Collins, Colorado **August 2009**
Master of Computer Science

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia **May 2004**
Bachelor of Science in Computer Engineering

ACADEMIC POSITIONS

Associate Professor of Computer Science & Engineering *August 2024 - present*
School of Engineering, Texas A&M International University

Associate Director *August 2024 - present*
Artificial Intelligence Center at Texas A&M International University, funded by NSF ExpandAI

Assistant Professor of Computer Science & Engineering *August 2018 - July 2024*
School of Engineering, Texas A&M International University

Visiting Faculty Researcher *June 2023 - August 2023*
Advanced Computing Systems Research, Oak Ridge National Laboratory

Visiting Scholar *June 2022 - August 2022*
Department of Computer Information Systems, Colorado State University

Lecturer *September 2016 - June 2017*
Department of Computer Science and Software Engineering, Seattle University

RESEARCH INTERESTS

- Software Engineering
- Artificial Intelligence & Machine Learning
- Model-Driven Software Development
- Cybersecurity
- High Performance Computing
- Software Verification and Validation

AWARDS, SCHOLARSHIPS AND HONORS

2025 Instructional Technology Excellence Award
Texas A&M International University

2024 and 2025 Distinguished Teacher of the Year Awards
College of Arts and Sciences, Texas A&M International University

2022 and 2023 Distinguished Scholar of the Year 2023 Awards
School of Engineering, Texas A&M International University

Doctoral Degree Scholarship
The Ministry of Higher Education, of Saudi Arabia

Master Degree Scholarship
The Ministry of Higher Education, Kingdom of Saudi Arabia

Gamma Beta Phi Honor Society
Colorado State University, Fort Collins, CO

UPE International Honor Society for the Computing Sciences
Colorado State University, Fort Collins, CO

MENTORING AND COLLABORATION

- Graduated Advisees and Co-advisees: 2 (M.S.), 39 (B.S.)
- Current Advisees and Co-advisees: 1 (Ph.D.), 1 (M.S.), 3 (B.S.)
- Collaborators count: 29 (DBLP)

TEACHING EXCELLENCE CERTIFICATIONS

Graduate Teaching Certificate *August 2011 - May 2016*
Institute for Learning and Teaching (TILT) at Colorado State University

Certificate in Effective College Instruction *August 2020 - May 2021*
Association of College and University Educators and the American Council on Education

Certificate in Teaching with Artificial Intelligence *Feb. 2024 - August 2024*
Sponsored by the Council on Academic Technology and Innovative Education

Teaching Online- An Introduction to Online Delivery *May 2020*
Quality Matters

Advanced Teaching Techniques: Level 2 *March 2020*
Top Hat

Miscellaneous Teaching Development and Training *August 2018 - present*
Numerous teaching seminars and workshops offered by TAMIU's Office of Information Technology Professional Development and Center for Faculty Development at Seattle University.

PUBLICATIONS

Notes:

1. The names of advisees are underlined and the names of international collaborators are in *italic font* and designated with asterisks *.

Refereed Journal Articles

1. Javaid, H., Petrescu, C.C., Schmunk, L.J., Monahan, J.M., O'Reilly, P., Garg, M., McGirr, L., Khasawneh, M.T., **Al Lail, M.**, Ganta, D. and Stubbs, T.M., 2025. The impact of artificial intelligence on biomarker discovery. *Emerging Topics in Life Sciences*, 8(2), pp.89-105.
2. Antonio Rosales and **Mustafa Al Lail**, "Taming the Frame Problem– An Automated Approach for Robust UML Class Diagram Specification and Verification". *Innovations in Systems and Software Engineering*, 1-23.

3. **Mustafa Al Lail**, Cervantes, Erick, Khaled Enab, Edwin Torres, and Khaled Almustafa. “Predicting Shale Oil Production in Real Time Using Supervised Machine Learning.” Accepted for publication in the International Scientific Journal Engineering Applications of Artificial Intelligence.
4. **Mustafa Al Lail**, Marshal Moncivais, Robert Benton, and *Alfredo Perez**. “Cloud-Based Access Control Including Time and Location.” *Electronics*, 13(14), 2812.
5. **Al Lail, Mustafa**, Alejandro Garcia, and Saul Olivo. “Machine Learning for Network Intrusion Detection—A Comparative Study.” *Future Internet* 15, no. 7 (2023): 243.
6. Ramadan Abdulnabi, **Mustafa Al-Lail**, Indrakshi Ray, Robert B. France, “Specification, Validation, and Enforcement of a Generalized Spatio-Temporal Role-Based Access Control Model” *IEEE Systems Journal* 7(3), 501-515 (2013)
7. **Al Lail, Mustafa**, Marshal Moncivais, and Miguelangel Trevino, “Towards a Software System for Spatio-Temporal Authorization,” *The Journal of Computing Sciences in Colleges*, (36, 7 2021).
8. Miguelangel Trevino and **Mustafa Al Lail**. “Database Web Application for Administering Spatio-Temporal Access Control Policies”. *American Journal of Advanced Research* 5 (2021): 1.

Refereed Conferences and Workshop Proceedings

1. **Huante, Alexis**, and Mustafa Al Lail. “Beyond Distance Metrics: A High-Performance Topological Classifier via Local Intrinsic Dimensionality.” Submitted to the Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026).
2. Trevino, Adriana, Mariana Garcia, Rodolfo Solis, and **Mustafa Al Lail**. “A Three-Tier Multi-modal AI Architecture for Human-Centric Safety in Collaborative Warehouse Environments.” To appear in the proceedings of The 2026 International Conference on Applied Computing: Bridging Theory, Innovation, and Real-World Impact (CAC 2026).
3. **Al Lail, Mustafa**. “Optimizing IoT Botnet Detection: A Comparative Study of Machine Learning and Feature Engineering Strategies.” In 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA), pp. 1-5. IEEE, 2025.
4. Moncivais, Marshal, and **Mustafa Al Lail**. “A Secure, Context-Aware Software Architecture for Real-Time IoT Applications’.” In 2025 28th International Symposium on Real-Time Distributed Computing (ISORC), pp. 303-309. IEEE, 2025..
5. **Mustafa Al Lail**, Huante, Alexis, and *Mariem Belhor**. “Impact of Network Data Complexity on Machine Learning Performance for Real-Time Iot Systems.” In 2025 28th International Symposium on Real-Time Distributed Computing (ISORC), pp. 1-8. IEEE, 2025.
6. Vela, Luis, Andy De Luna, and **Mustafa Al Lail**. “Integrating Collaborative Ground-Aerial Robotics for Enhanced Safety and System Optimization in Automated Warehouse Environments.” To appear in the proceedings of 2025 AHFE Hawaii International Conference.
7. **Al Lail, Mustafa**, Pinto, Daniela, Almanza, Luis Alvarez, Salazar, Francisco, and Rizzi, Carolina. “Beyond Traditional Methods: Deep Learning with Data Augmentation for Robust Access Control.” In 2024 33rd International Conference on Computer Communications and Networks (ICCCN), pp. 1-6. IEEE, 2024.
8. Viesca, Antonio Rosales, **Mustafa Al Lail**, and *Omar Alam**. “Streamlining cps validation: Using interoperable uml tools for seamless model exchange.” In 2024 IEEE 27th International Symposium on Real-Time Distributed Computing (ISORC), pp. 1-4. IEEE, 2024.

9. **Al Lail, Mustafa**, Antonio Rosales, Hector Cardenas, *Mohammad Zarour**, and *Alfredo Perez**, “TPV: A Tool for Validating Temporal Properties in UML Class Diagrams”. In Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings, pp. 114-118. 2024.
10. *Pedro Valero – Lara**, Alexis Huante, **Mustafa Al Lail**, *William Godoy**, *Keita Teranishi**, *Prasanna Balaprakash**, and *Jeffrey Vetter**. “Comparing Llama-2 and GPT-3 LLMs for HPC kernels generation”. Proceedings of the 36th International Workshop on Languages and Compilers for Parallel Computing (LCPC2023).
11. Hector Cardenas, and **Mustafa Al Lail**, “Specifying Temporal Properties in UML Using Patterns: A Tool-supported Approach”. Proceedings of the ACM/IEEE 26th International Conference on Model-Driven Engineering Languages and Systems (MODELS 2023).
12. Antonio Rosales, and **Mustafa Al Lail**, “Automated Mitigation of Frame Problem in UML Class Diagram Verification”. Proceedings of the ACM/IEEE 26th International Conference on Model-Driven Engineering Languages and Systems (MODELS 2023).
13. Moncivais, Marshal, and **Mustafa Al Lail**. “Poster: Integrating Spatio-temporal Authorization with Generic Cloud-based Software Architecture for Internet of Things Devices.” Proceedings of the 28th ACM Symposium on Access Control Models and Technologies, pp. 59-61. 2023.
14. Cervantes, Erick, **Mustafa Al Lail**, Edwin Torres, and Khaled Enab. “Performance Evaluation of Supervised Machine Learning Algorithms to Predict Shale Oil Production..” In IISE Annual Conference and Expo. IISE, 2023.
15. *Lars Hamann**, *Martin Gogolla**, and **Mustafa Al Lail**. “Categorization of Approaches to Extend and Reuse OCL”. Proceedings of the 25th International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, pp. 847-851. 2022.
16. **Al Lail, Mustafa**, Antonio Rosales, Hector Cardenas, *Lars Hamann**, and *Alfredo Perez**. “Transformation of TOCL Temporal Properties into OCL”. Proceedings of the 25th International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings, pp. 847-851. 2022.
17. Hector Cardenas, Ryan Zimmerman, Antonio Rosales, **Mustafa Al Lail**, and *Alfredo Perez**, “Formal UML-based Modeling and Analysis for Securing Location-based IoT Applications”. Proceedings of the IEEE 19th International Conference on Mobile Ad Hoc and Smart Systems (MASS), pp. 722-723. IEEE, 2022.
18. **Al Lail, Mustafa**, Alejandro Garcia, and Saul Olivo, “Machine Learning for Network Intrusion Detection- A Comparative Study”. Presented at IISE Lean Six Sigma & Data Science Conference 2022. Atlanta, Georgia, September 19 – 21, 2022.
19. **Mustafa Al Lail**, “Towards Cloud-Based Software for Incorporating Time and Location Into Access Control Decisions”, In Proceedings of the 26th ACM Symposium on Access Control Models and Technologies, pp. 55-57. 2021.
20. **Mustafa Al-Lail**, Wuliang Sun, Robert B. France, “Analyzing Behavioral Aspects of UML Design Class Models Against Temporal Properties”, In Proceedings of The 14th International Conference on Quality Software (QSIC 2014), Dallas, Texas, USA October 2014, 196-201
21. **Mustafa Al-Lail**, Ramadan Abdunabi, Robert B. France, Indrakshi Ray : “An Approach to Analyzing Temporal Properties in UML Class Models”, in Proceedings of MoDeV@MoDELS 2013: 77-86
22. **Mustafa Al-Lail**, “A Framework for Specifying and Analyzing Temporal Properties of UML Class Models”. Demos/Posters/StudentResearch@MoDELS 2013: 112-117

23. **Mustafa Al-Lail**, Ramadan Abdunabi, Robert B France, Indrakshi Ray, 2013, “Rigorous Analysis of Temporal Access Control Properties in Mobile Systems”, in Proceedings of 18th IEEE International Conference on the Engineering of Complex Computer Systems, Singapore July 2013, 246-251

TEACHING EXPERIENCE AND INTERESTS

Graduate Courses

- Object Oriented Analysis and Design
- Software Architecture
- Software Design Patterns
- Internet of Things (IoT) and Cloud Computing
- Applied Data Mining
- Cybersecurity
- Software Verification and Validation

Undergraduate Courses

- Software Development
- Machine Learning
- Computer Security
- Software Engineering and Project Development
- Fundamentals of Databases
- Networks and Data Communication
- Undergraduate Research

GRANTS

Funded Projects

1. PI: Khaled Enab, **Co-PI: Mustafa Al Lail**, Co-PI: Deepak Ganta, and Co-PI: Kenneth Tobin. “AI Institute: Physics-Based AI for Energy Transition”, National Science Foundation (NSF), \$399,855.
2. PI: Kenneth Tobin, **Co-PI: Mustafa Al Lail**, Co-PI: Deepak Ganta, and Co-PI: Kenneth Tobin. “STEM Connector”, The National Aeronautics and Space Administration (NASA), \$963,000.
3. PD: Alfredo Ramirez, Co-PD:Kameron Jorgensen, Senior Faculty: **Mustafa Al Lail**, “Academic Recovery and Data Analysis (ARDA)”, USA Department of Education, \$2,999,783.00.
4. **PI: Mustafa Al Lail**, “Class Diagram Analyzer (CDA): Tackling the Software Crisis by the Integration of Formal Methods and Model-Driven Engineering Through the Development of a Novel, Effective, Efficient, and Usable Tool ”, Texas A&M International University Presidential Research Development Grant, 09/23 - 02/25, \$50,000.
5. **PI: Mustafa Al Lail**, “Building Software and Data Analytic Applications for Solving DOE Scientific Problems”, USA Department of Energy, 06/23 - 08/23, \$49,800.
6. **PI: Mustafa Al Lail**, “Live Parking Lot Maps using Computer Vision”, Texas A&M International University, 09/23 - 05/24, \$44,000.
7. **PI: Mustafa Al Lail**, “Building Heart Rhythms Classification Device for Human”, Texas A&M International University, 12/22 - 05/23, \$11,500.
8. **PI: Mustafa Al Lail**, “Addressing the Complexity of Access Control Decision and Administration Through the Use of Artificial Intelligence Techniques”, Texas A&M International University, 12/22 - 05/23, \$6,000.

9. **PI: Mustafa Al Lail**, "Investigating the Influence of Dataset Complexity on the Performance of Machine Learning Models for Network Intrusion Detection", Texas A&M International University, 08/23 - 05/24, \$6,000.
10. **PI: Mustafa Al Lail**, "Phishing Detection Using Machine Learning", Texas A&M International University, 01/22 - 05/24, \$6,000.
11. **PI: Mustafa Al Lail**, "Improvements of The Tool Support for Uncovering Faults in Software Design Models", University Research Grant, Texas A&M International University, 9/22 - 8/23, \$10,000.
12. **PI: Mustafa Al Lail**, "A Tool Support for Uncovering Faults in Software Design Models", University Research Grant, Texas A&M International University, 9/21 - 8/22, \$10,000.
13. **PI: Mustafa Al Lail**, "Secure Cloud-based Software System Incorporating Time and Location Into Access Control Decisions", ACT on IDEAS Fellowship Mini Grant, Texas A&M International University, 1/22 - 5/22, \$3,000.
14. **PI: Mustafa Al Lail**, "Towards Cloud-Based Software for Incorporating Time and Location Into Access Control Decisions", ACT on IDEAS Fellowship Mini Grant, Texas A&M International University, 8/21 - 12/21, \$3,000.
15. **PI: Mustafa Al Lail**, "Software Implementation of Generalized Spatio-Temporal Role-Based Access Control Model (GSTRBAC)", ACT on IDEAS Fellowship Mini Grant, Texas A&M International University, 1/21 - 5/21, \$3,000.
16. **PI: Mustafa Al Lail**, "Software Implementation of Generalized Spatio-Temporal Role-Based Access Control Model (GSTRBAC)", ACT on IDEAS Fellowship Mini Grant, Texas A&M International University, 8/20 - 12/20, \$3,000.

STUDENT SUPERVISION

Undergraduate Research Conference Oral and Poster Presentations

1. Miguelangel Trevino, Mustafa Al Lail, "Database Web Application for Administering Spatio-Temporal Access Control Policies". Accepted at the 2021 International Conference of Advanced Research in Applied Science, Engineering and Technology (ICARASET'21).
2. Antonio Rosales, Hector Cardenas and Mustafa Al Lail,, "A Tool Support for Uncovering Faults in Software Design Models". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
3. Edwin Torres, Erick Cervantes, Mustafa Al Lail, and Khaled Enab, "Efficient Prediction of Shale Oil Production Rates Using Machine Learning". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
4. Marshal Moncivais, Carlos Delgado, Robert Benton, Mustafa Al Lail, and Alfredo Perez, "Secure Cloud-based Software System Incorporating Time and Location Into Access Control Decisions". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
5. Jesus Pachicano, David Haunschmied, and Mustafa Al Lail, "Detecting Network Intrusions Using Supervised Machine Learning Techniques and Feature Selection". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
6. Steven Rangel, Sofia Garza, and Mustafa Al Lail, "Application of Machine Learning Algorithms for Intrusion Attacks". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.

7. Alejandro Garcia, Saul Olivo, and Mustafa Al Lail, "Machine Learning Approach to Intrusion Detection ". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
8. Mario Juarez, Carlos Gonzalez, and Mustafa Al Lail, "Machine Learning Technique for Intrusion Detection in Cloud Computing and Internet of Things". Poster presentation at the 17th Annual PATHWAYS Student Research Symposium, College Station, TX, March 3-4, 2022.
9. Marshal Moncivais, Miguelangel Trevino, Carlos Delgado, and Mustafa Al Lail, "Software for Incorporating Time and Location into Access Control Decisions". Poster presentation at the 2021 Texas Undergraduate Research Day Conference, Feb. 23-24, 2021.
10. Omar Botello, Reynaldo Vielma, Marshal Moncivais, Jaqueline Vega, Miguelangel Trevino, and Mustafa Al Lail, "A Software Design for Mobile Applications with Spatio-Temporal Authorization," Poster presentation at the 16th Annual PATHWAYS Student Research Symposium, Laredo, TX, Nov. 7-8, 2019.
11. Alex Aleman and Mustafa Al Lail, "An Implementation of GSTRBAC - A Novel Access Control Model for Mobile Applications". Poster presentation at Lamar Bruni Vergara Academic Conference 2019, Laredo, TX.
12. Hector Cardenas and Mustafa Al Lail,, "A Tool Support for Uncovering Faults in Software Design Models". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
13. Jacob Flores, Javid Alasgarli, and Mustafa Al Lail,, "Preemptive IoT botnet detection using machine learning model". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
14. Erick Cervantes, Jacob Flores, and Mustafa Al Lail,, "Domain Name-based Dataset for Detecting Phishing Attacks ". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
15. Mario Juarez, Luis Moreno, Felipe Rivas and Mustafa Al Lail,, "Building Heart Rhythms Classification Device for Humans". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
16. Luis Moreno, Daniel Salinas and Mustafa Al Lail,, "Machine Learning Based Detection on DDos Attacks for Iot Systems". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
17. Alexis Haunte, Saul Olivo, Vilen Elliott and Mustafa Al Lail,, "Intrinsic Dimensionality for Machine Learning - A Comparative Study". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.
18. Jhonny Gonzalez, Jose Gonzalez and Mustafa Al Lail,, "Creating a Graphical User Interface to Design UML Class Diagrams for Analysis". Poster presentation at the 18th Annual PATHWAYS Student Research Symposium, Galveston, TX, March 2-3, 2023.

Undergraduate Capstone Project

1. David Hatt, Matthew Wenala, Yousef Alturaifi, Zean Rivera (September 2016 - June 2017)
 Project Title: "PACCAR BOM FEEDBACK "
 Software delivered to PACCAR Inc. company
 Poster presented at Seattle University's Project Day event, Seattle, WA

INDUSTRY POSITIONS

Khalid Ali Alturki & Sons*Senior Network Engineer*

November 2004 - February 2006

Alkhobar, Saudi Arabia

- Designed and implemented a WAN project for connecting ten sites to the company's corporate network.
- Installed, configured, and maintained the servers and services for the above project such as: DNS servers, exchange servers, DHCP, ISA, SQL, file/print, and Symantec Anti-Virus solution.
- Performed network, server, PC, and users troubleshooting to isolate and resolve problems.
- Designed and supervised a structured wiring project as project manager.
- Trained and supervised a group of IT specialists.
- Monitored various hardware and software platforms on a regular basis to ensure efficiency.
- Installed and configured wireless network services.
- Participated in managing many IT hardware, software, and network projects.
- Implemented the necessary changes for upgrading hardware and software network components.

Amaleed Trading Est*Network Administrator*

January 2001- October 2004

Alkhobar, Saudi Arabia

- Maintained, upgraded, and installed service packs on Windows 2000 servers.
- Performed Active Directory and Exchange tasks such as creating new users' accounts and emails.
- Provided network, server, PCs, and users troubleshooting to isolate and resolve common problems.
- Assisted the IT technical in providing users services by addressing users' requests during peak periods.
- Evaluated, recommended and applied new technology and solutions to optimize operating efficiency.
- Installed, upgraded, and supported desktop applications on Windows 2000 and XP.
- Performed escalated users support.

King Fahd University of Petroleum and Minerals*Student Network Administrator*

February 2003-January 2004

Dhahran, Saudi Arabia

- Secured, maintained, and replaced laboratory equipment, supplies, and personal computers.
- Coordinated system administration tasks including creating students' accounts and email addresses.
- Documented all network procedures of existing systems.
- Provided technical support for students and faculty.
- Installed, upgraded, and configured network printers.

International Business Machines*Network Engineer Intern*

June 2002 - August 2002

Alkhobar, Saudi Arabia

- Installed and configured an IP/TV network service at King Fahd University of Petroleum and Minerals (KFUPM).
- Installed and configured a wireless network at KFUPM.
- Participated in design of Marafiq company's network infrastructure.
- Simulated and implemented File/Print server for Marafiq Company project.

GRADUATE SCHOOL POSITIONS

Dept. of Computer Science, Colorado State University

January 2016 - May 2016

Graduate Teaching Assistant

Fort Collins, CO

- Assisted in teaching CS470: Computer Architecture.
- Prepared and graded quizzes, homework assignments, and midterm exams.
- Prepared and presented a number of lectures.

Dept. of Computer Science, Colorado State University

August 2009- May 2016

Guest Lecturer

Fort Collins, CO

- Prepared and presented a number of guest lectures in the security and software engineering classes at Colorado State University such as: CS 556 Computer Security, CS517 Software Specification & Design, CS414 Object Oriented Design, and CS314 Software Engineering.
- Prepared and presented numerous lectures and talks in the Security and Software Engineering research groups at Colorado State University.

School of Global Environmental Sustainability, Colorado State University

April

2015-August 2015

Graduate Research Assistant

Fort Collins, CO

- Created a website geared towards analyzing and curating time-lapse imagery of rivers using the PyBossa crowdsource framework.

Dept. of Computer Science, Colorado State University

August 2011-May 2014

Graduate Research Assistant

Fort Collins, CO

- Carried out extensive research activities on three projects:
 1. Specification and Analysis of Access Control Models for Mobile Applications
 2. A Framework for Specifying and Analyzing Temporal Properties
 3. Lightweight Analysis of Software Models

Dept. of Philosophy, Colorado State University

August 2008 - December 2008

Graduate Teaching Assistant

Fort Collins, CO

- Assisted in teaching and grading a graduate course Classical Arabic 1: Advanced Introduction.
- Carried out the usual duties of teaching assistant such as grading, helping student to understand the course material, and test and course preparation.
- Prepared and presented several lectures and review sessions.

Dept. of Philosophy, Colorado State University

May 2008 - August 2008

Graduate Research Assistant

Fort Collins, CO

- Helped in the development of a new computer-software system for academic and scholarly typesetting and typography. The Oriental TEX project was initiated to facilitate the development of high quality typography and typesetting of academic and scholarly texts that require the Arabic script, such as critical editions and monographs.

PROFESSIONAL/LEADERSHIP SERVICES AND ACTIVITIES

Scientific Community Service

Program Committee Co-Chair

The 28th IEEE International Symposium on Real-Time Distributed Computing, Toulouse, France, 2025.

Steering Committee Member

Consortium for Computing Sciences in Colleges (CCSC): South Central Region April 2021 - present

Leadership Training

TAMU's Leadership Experience Program

August 2019 - May 2020

The purpose of this program is to provide participants with knowledge, skills, tools, and connections to effectively lead people and organizations.

Professional Affiliations

Association for Computing Machinery (ACM), Institute of Electrical and Electronics Engineers (IEEE), Upsilon Pi Epsilon (UPE)

Review Activities

- ACM-IEEE International Conference on Software Engineering (ICSE)
- ACM-IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS)
- Workshop on Modeling in Software Engineering at ICSE (MiSE)
- ACM-IEEE International Conference on Automated Software Engineering (ASE)
- International Conference Software Language Engineering (SLE)
- ACM-IEEE Symposium on Formal Methods and Models for System Design
- Journal of Software and System Modeling (SoSyM)
- Journal of IEEE Transactions on Software Engineering (TSE)

Student Volunteer at Conferences

- 13th ACM Symposium on Access Control Models and Technologies, Estes Park, CO
- ACM/IEEE 16th International Conference on Model Driven Engineering Languages and Systems, Denver, CO
- Workshop on Modeling in Software Engineering at ICSE 2009, Vancouver, Canada

University, College, Department Service

Committees

- School of Engineering search committee for Endowed Full-Professor of Computer Engineering, Spring 2022
- School of Engineering search committee for Visiting Associate/Assistant Professor of Computer Engineering, Spring 2022

- School of Engineering search committee for Petroleum Engineering Assistant professor, Spring 2020
- ABET accreditation committee of System Engineering, Spring 2019
- Assessment committee of the B.S of Computer Engineering and the Computer Science minor
- B.S. Computer Engineering Curriculum committee
- Computer Science Minor Curriculum committee

Miscellaneous Service

Computer Science & Engineering Association at TAMIU: I have established and advised the Computer Science club at Texas A&M International University since Fall 2019. The club aims to build and sustain a community of students to be part of in what can often look like an isolated field of study at TAMIU.

Computer Science Tutoring Program: I have established a new tutoring program for Computer Science. This program aims to support computer science education and attract more students to computer science and engineering majors.

Conference Judging

- 17th Annual PATHWAYS Student Research Symposium, March, 2022
- United ISD's Annual Science Fair Science Fair 2020
- The 2019 LBV Academic Conference, April, 2019
- 16th Annual PATHWAYS Student Research Symposium, November, 2019
- The School of Engineering and Math Conference at TAMIU, November, 2019.
- The School of Engineering and Math Conference at TAMIU, November, 2018.

Student Recruitment Activities

- Computer Science and Engineering Q&A Session
- Virtual Student Organization Fair
- Halloween Fest at TAMIU.
- Movie Night.

Community Service

- 2019-2020 STEM Advisory Board for Harmony Public Schools

Upsilon Pi Epsilon International Honor Society for the Computing Sciences Beta Chapter at Colorado State University

- President (2013-2015)
- Vice-President (2012-2013)
- Treasurer (2011-2012)
- Historian (2010-2011)
- Member (2010-present)

Athletic Activities

- Kingdom of Saudi Arabia Waterpolo National Team 1995-2001

- Al Safa Club Swimming and Waterpolo teams 1991-2002
- Al Qadisiyah Waterpolo Team 2002-2004
- King Fahd University Swimming Team 1997-2003
- Colorado State University Waterpolo Team 2015-2016